§5 Standard operation form and inter-time connection — part 2 of 2

Reference: subsec:s5_intertime, subsec:s5_internal_quantities, subsec:s5_summary

Source: paper/section5_extended_dynamics.tex, lines 294–626 (inter-time change induced by a carrier; classification-C quantities: bias and bias strength, readout types, prediction accuracy coefficient and effective time horizon, effective development freedom, PSL and ISL health, the organizing tuple; hand-off to later sections), at commit e6e12cd. Status: draft. Part 1 is 05_dynamics_1.md; §6 begins in 06_relational_1.md.

Inter-time change

Reference: subsec:s5_intertime, subsubsec:intertime_update

Definition (def:intertime_update): inter-time change induced from a carrier. Specify a non-empty set D_{rd} and a fixed readout $\mathrm{rd} : X \times \mathcal{M} \times E_{\mathrm{ext}} \to \mathrm{D}_{\mathrm{rd}}$. With $\mathrm{rd}_t = \mathrm{rd}(x_t, M_t, e_t)$, the next-time value is $\mathrm{rd}_{t+1} = \mathrm{rd}(f(x_t, u_t, e_t), \mathrm{Update}(M_t, x_t, e_t), e_{t+1})$ with $e_{t+1} = (b_{t+1}, g_O(o_t, x_t, u_t, b_t))$. This type preservation needs no independent update map. When e_{t+1} is required, it is not taken to be an internal prediction at time t. A quantity that takes a trajectory as its argument is evaluated from that trajectory and the specified readout rule.

Definition (def:learning_intertime_change): inter-time position of learning and forgetting (classification B). A change of persistent content involved in learning or forgetting is described as a change of M_t^{L}. The change is not identified with improvement or deterioration. A change of the working component alone is not called learning in this definition. Learning stop on a sequence follows def:update_stop_on_trajectory. The change quantities $\delta_{M,c,t}, \delta_t, \lambda_{T,t}$ remain quantities that include the working component.

High-speed processing, Automaticity and change of state (rem:s5_fast_change). The operating forms of §4 can form and change with changes of the learning component, but their holding is not a function of the learning component alone: the working component, the current input, and the states of base and sensitivity can also be involved. This paper does not state as a general consequence that the invariance of high-speed processing or Automaticity follows from learning stop, and it places no independent formation speed or learning rule. Time changes of sensitivity, base, effectors and health are described through the transition of x_t.

Learning and the feasibility structure (rem:s5_learning_viability). Learning is a change of M_t^{L}, while the 7-tuple does not depend on time (the convention on levels of description in §2). Therefore K, the feasible-action correspondence A, $\mathrm{Viab}_f(K)$ and the kernel-maintaining response set U^{Viab} (def:kernel_response) are not changed by learning. What learning can affect is which effective action is selected on the operation record, in particular whether $u_t \in U^{\mathrm{Viab}}(x_t)$ holds (def:kernel_response_acquisition), and the trajectory through it. Bodily change, being a change of state, is described through f as a transition of x_t and does not change these sets. This paper does not claim that learning leads to the acquisition of kernel-maintaining responses (rem:kernel_response_scope).

Classification-C quantities

Reference: subsec:s5_internal_quantities

Position of bias and bias strength

Reference: subsec:beta_dyn

Definition (def:bias_tendency): bias (classification C). A *bias* is a tendency of one deviation, relative to a specified reference standard, in carrying out the responsibility of some division of the primitive functions (def:primitive_functions); carrying out includes the allocation or use of processing resources, attention and the like. A non-empty index set $\mathcal{J}_\beta$, fixed over time and not assumed finite, indexes the biases considered; each $\iota \in \mathcal{J}_\beta$ indexes one bias. The cutting of biases by $\mathcal{J}_\beta$ is relative to the resolution of description; finer or coarser divisions are not excluded. Coexistence of several biases and their having different directions are not excluded. The index is an identifier and does not express direction or content. The reference standard is an agreement on the state with no deviation in the responsibility concerned and is not a normative ideal, a correct performance or an optimal performance; no reference standard common to all objects is fixed. A comparison of strength over time for the same index fixes the reference standard and the scale over the period.

Definition (def:bias_attribution): attribution division of a bias (classification C). With $\mathrm{Prim} = \{\mathrm{sens}, \mathrm{disc}, \mathrm{val}, \mathrm{act}\}$ (sensing, discrimination, evaluation, action; matched with the indices of the standard functional forms of def:three_primitives), the *attribution map* $\mathrm{loc}_\beta : \mathcal{J}_\beta \to \mathcal{P}(\mathrm{Prim}) \setminus \{\emptyset\}$ gives each bias a non-empty set of divisions where the deviation appears, its *attribution division*. A bias is attributed to a single division if $|\mathrm{loc}_\beta(\iota)| = 1$ and compositely if $|\mathrm{loc}_\beta(\iota)| \geq 2$. The attribution division is fixed over time, and the reference standard of def:bias_tendency is specified in correspondence with it.

Status of the attribution division (rem:bias_attribution_status). The attribution division is what the theory specifies, per object of description, as the division in which a bias is described; it does not require the ACS to represent or grasp its own biases internally (subsec:description_levels_global). This paper does not claim that observed behavior determines the attribution division uniquely, because the same deviation of behavior is compatible with descriptions that attribute it to different divisions. The manuscript notes that in studies of perceptual decision-making an observed choice bias is explained, depending on its kind and conditions, either as a change of the starting point of evidence accumulation or as a change of its process, and that this is discriminated by fitting and comparing diffusion models (Mulder et al. 2012; Urai et al. 2019); it also cites an argument that emphasis should be placed on building and testing observer models (Rahnev and Denison 2018). There are $2^4 - 1 = 15$ non-empty subsets of divisions; they are used only as a descriptive classification, and no regularity that particular combinations are likely to appear is claimed.

Definition (def:beta_accumulated): bias strength (classification C). For $\iota \in \mathcal{J}_\beta$ and time t, the non-negative quantity $\beta_{\iota,t}$ is the degree of deviation of bias ι from its reference standard at time t, the *bias strength*. $\beta_{\iota,t} = 0$ means no deviation from the reference standard at t. The scale is specified per bias, and comparability of values across different biases is not assumed. The family $\boldsymbol{\beta}_t = (\beta_{\iota,t})_{\iota \in \mathcal{J}_\beta}$ is not an aggregation into one real number. It is defined as non-negative reals here, and the bounded readout type used follows def:s5_readout_types. When the reference standard or the scale is unspecified, or the strength is unmeasured or unidentified, $\beta_{\iota,t} = 0$ is not assigned for that reason alone. $\beta_{\iota,t}$ expresses degree, not direction or content.

Bias strength is a quantity describing biases that appear in the use of capability and resources, not capability itself. As in rem:cognitive_capacity_policy, this paper does not include bias strength in cognitive processing capability. Formalizing the direction and content of biases is a Phase 2 task (L7 of subsec:limitations_list).

Abstract structural description versus implementation description (rem:beta_multiple_biases). def:bias_tendency and def:beta_accumulated fix bias and bias strength at the level of abstract structural description, where several biases can exist, each with a strength. At the level of implementation description, a bias and its strength can change with the content of

updates and can be integrated with other biases, whether the bias is formed by learning or given by implementation or design. This paper does not fix over time which implementation content corresponds to which index ι. When the change or integration is expressed at the abstract level, it is described as a change of the state or internal model that carries the readout of `def:beta_intertime_update`; no independent update map is placed on the family of strengths. This paper places no general formula aggregating several strengths into one real number and does not assume that biases of different directions cancel or reinforce one another. The correspondence of implementation biases to indices, the representation of direction and content, and the formalization of relations among biases are Phase 2 tasks (L7).

Formation versus attribution of a bias (`rem:bias_formation_vs_attribution`). A bias can be formed by any of the three mechanisms M1, M2, M3 of §4, by PSL or ISL, or by their joint action. This paper distinguishes which mechanism forms a bias from which division it is attributed to (`def:bias_attribution`) and places no one-to-one correspondence between them, as in §2 where no one-to-one correspondence between TIPA mechanisms and primitive-function divisions is placed. A deviation arising from the states of PSL and ISL related to the base functions (`def:base_functions`) is also described by attributing it to the division where it appears. Description of formation per mechanism, the correspondence between combinations of formation and attribution divisions, and identification of attribution divisions are tasks of Phase 1.α or Phase 2 (L7, `subsec:phase1a`).

Value of bias strength is not a judgment of quality (`rem:beta_value_not_quality`). This paper does not read a larger or smaller $\beta_{\iota,t}$ as improvement, deterioration, error or irrationality. How a bias bears on the feasibility of the ACS depends on its content and on the environment and internal state in which it acts, so it is not determined by the value alone. Increase, maintenance and decrease of $\beta_{\iota,t}$ are all allowed, and fixing its value is not identified with a bias accumulating in one direction. This paper does not state, from an increase or decrease of $\beta_{\iota,t}$, the direction of change of prediction accuracy as a general consequence.

Bias strength versus state estimation error (`rem:beta_vs_estimation_error`). Bias strength is the degree of deviation in carrying out a primitive function relative to the reference standard of `def:bias_tendency`. The state estimation error $\varepsilon_{\mathrm{est},t}$ (`def:s4_state_estimation_error`) is a gap between the internal estimate and the theoretical state, and differs from bias strength in the object referred to and the scale. This paper places no general implication in either direction between positive bias strength and positive state estimation error. The distinction does not claim that there is no relation between them or that they are statistically independent.

Readout types and inter-time readout of bias strength

Definition (`def:s5_readout_types`): readout types and analysis range. Fix finite $\beta_{\max} > 0$, $D_{\max} > 0$ and a prediction depth $\tau \in \mathbb{N}$, $\tau \geq 1$. With $\mathcal{H} = \{\mathrm{psl}, \mathrm{mem}, \mathrm{sens}, \mathrm{eff}\}$, the description rules adopted in this section are the total maps $\mathrm{rd}_{\beta,\iota} : X \times \mathcal{M} \times E_{\mathrm{ext}} \to [0, \beta_{\max}]$ ($\iota \in \partial_\beta$), $\mathrm{rd}_{h,q} : X \to [0,1]$ ($q \in \mathcal{H}$), $\mathrm{rd}_d : X \times \mathcal{M} \to [0, D_{\max}]$ and $\mathrm{rd}_{\pi,\tau} : (X \times \mathcal{M} \times E_{\mathrm{ext}})^{\tau+1} \to [0,1]$. $\mathrm{rd}_{\beta,\iota}$ is specified as a rule returning the strength corresponding to the reference standard of `def:bias_tendency` and the scale of `def:beta_accumulated`; each scale is normalized into $[0, \beta_{\max}]$ preserving the meaning of zero. This paper does not state comparability of different biases from the common range as a general consequence. These are types of description rules specified per object and not claims that derive concrete rules uniquely. The ordinary subspace topology is placed on each range but continuity of readouts is not imposed. The analysis range is descriptions that can be normalized to a finite upper bound; no claim is made that every phenomenon falls in it. This instantaneous readout is not applied to a quantity not determined by the present carriers alone; a description requiring history is specified separately as a readout on the operation record without adding an independent state.

Definition (`def:beta_intertime_update`): inter-time readout of bias strength. Fixing the rules of `def:s5_readout_types`, $\beta_{\iota,t} = \mathrm{rd}_{\beta,\iota}(x_t, M_t, e_t)$. The product topology is placed on $[0, \beta_{\max}]^{\partial_\beta}$, which is compact because each factor is; this is not used as an independent state space or as the domain of an update map.

Prediction accuracy coefficient and effective time horizon

Reference: subsubsec:s5_prediction_coefficient, subsec:tau_hat_section5

Definition (def:s5_prediction_coefficient): prediction accuracy coefficient (classification C). For the state–environment subsequence $z_s = (x_s, M_s, e_s)$ of a standard operation record, $\pi_{\tau,t} = \mathrm{rd}_{\pi,\tau}(z_t, \dots, z_{t+\tau}) \in [0,1]$ is a coefficient describing after the fact the effectiveness of the prediction at time t. The adopted rule fixes the prediction target and the matching method per object. Its quantification, measurability and prediction success are not claimed in this section. It refers to future realized values and is not an internal input of the ACS at time t.

Definition (def:tau_hat_section5): effective time horizon (classification C). $\hat{\tau}_t := \tau \pi_{\tau,t} \in [0, \tau]$, so $\hat{\tau}_{t+1} - \hat{\tau}_t = \tau(\pi_{\tau,t+1} - \pi_{\tau,t})$. τ is a discrete prediction depth and $\hat{\tau}_t$ a real-valued scale; no state at non-integer times such as $x_{t+\hat{\tau}_t}$ is introduced.

Effective development freedom

Reference: subsec:D_eff_section5

Definition (def:deff_section5): effective development freedom (classification C). $d_{\mathrm{eff},t} := \mathrm{rd}_d(x_t, M_t) \in [0, D_{\max}]$ is a quantity describing the effective freedom of development. The range and carrier are specified, and the concrete computing rule is left to Phase 1.α. It differs from $|A_t|$, from the action diversity of M2 and from relation quality (subsec:ro_quality). This paper does not state, from its value, existence or increase or decrease of feasible actions as a general consequence.

PSL and ISL health

Reference: subsec:health_dyn

Definition (def:s5_health_quantities): PSL and ISL health quantities (classification C). $h_{q,t} := \mathrm{rd}_{h,q}(x_t) \in [0,1]$, where $q = \mathrm{psl}, \mathrm{mem}, \mathrm{sens}, \mathrm{eff}$ designate the health of the base, the memory implementation, sensing and action. Evaluation of learned content itself is distinguished from the health of the memory implementation. Common unit, mutual independence and direct comparability are not required.

Health, functional axis and diagnosis (rem:s5_health_noncollapse). Health and the functional axis of §4 are different descriptive quantities, and no general mutual determination is placed. Internal diagnosis of the ACS depends on available information and the state of the TIPA functions, and external diagnosis on observed information and measurement rules. This paper does not state, from $h_{q,t} = 0$, $d_{\mathrm{eff},t} = 0$ or $\pi_{\tau,t} = 0$ alone, blockage, falsity of the time-indexed satisfaction predicate of FR, functional termination or collapse as a general consequence.

Definition (def:extended_variable_tuple, subsubsec:desc_tuple): organizing tuple of time-dependent quantities (classification C). $\mathrm{Desc}_t = (\boldsymbol{\beta}_t, d_{\mathrm{eff},t}, \pi_{\tau,t}, \hat{\tau}_t, h_{\mathrm{psl},t}, h_{\mathrm{mem},t}, h_{\mathrm{sens},t}, h_{\mathrm{eff},t})$ is an octuple collecting the quantities read out and does not change the 7-tuple of the ACS. It is neither an independent state nor a bundle of same-time internally available information. Because it contains the prediction coefficient, it has the character of an after-the-fact evaluation of the record.

Hand-off to later sections

Reference: subsec:s5_summary, subsubsec:s5_deff_interface

Definition (def:s5_downstream_interface): record interface to later sections. To §6 (sec:rel_min) the sequences of $A_t, d_{\mathrm{eff},t}, \hat{\tau}_t$ are handed over; the per-individual indices and the finite-window differences follow §6. Evaluating a window containing the prediction coefficient also needs realized records beyond the window end and does not imply online diagnosability at the same time. To §7 (sec:collapse_cond) the standard operation record and Desc_t are handed over. The standard operation record is mapped to the trajectory record of §7 when the records of WF, FT, Col and other realization-level quantities needed there are given separately

and the consistency with the dynamics presupposed by the trajectory record (`rem:trajectory_space_scope`) is satisfied. P2 alone does not determine these indicators or recoverability under admissible external actions uniquely. Interpretation as the same ACS is limited to $I_{\mathrm{ACS}}(\sigma)$. When a prediction window straddling collapse is used as a prediction evaluation of the same ACS, it is confirmed that the whole window lies in the description domain of that ACS.

The only assumption of §5 is the observation factorization. P1–P3 rest on composition of fixed total maps and recursion on the natural numbers. Reaching in finite time, recovery, invariance, stability and implementability are separate proof obligations and are not claimed from the type preservation or the boundedness of the classification-C quantities of this section.

Check status. This summary adds no new definition, assumption or result. Labels in backticks are shared with the manuscript. It has been checked against the manuscript by the drafting model and audited by an external AI model. It remains a draft until the author checks it against the manuscript. No independent human third-party verification is claimed.

References

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